Use-Case Flow Specification

Airport Lost Luggage Claim & Tracking Portal — Problem Statement #30

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Use Case ID / Name: UC-01 — Register Lost Baggage Claim

Primary Actor: Passenger

Supporting Actors: Airport Scan System (via UC-05 Match Lost Baggage), Notification Service

Related Requirements: FR-001, FR-002, NFR-002

Preconditions

1. The Passenger has successfully completed UC-08 Authenticate User (an «include» relationship — the system verifies passenger identity before any claim data is accessed or stored).
2. The Passenger's flight has landed and baggage has not been collected at the carousel.
3. The Airport Scan System is online and reachable so scan logs can be cross-referenced.

Postconditions

Success: A new baggage claim record is created with a unique Claim ID and status "Received"; the claim is queued for automatic matching against unrouted scan logs (UC-05).

Failure: No claim record is created, and the Passenger is informed of the specific reason for rejection.

Main Success Scenario

1. Passenger selects "Register Lost Baggage Claim" from the portal home screen.

2. System displays the claim form requesting: baggage tag number, flight number, travel date, and contact details.

3. Passenger enters the requested details and submits the form.

4. System validates that all mandatory fields are present and that the baggage tag matches the required tag format (FR-002).

5. System cross-references the baggage tag against unrouted luggage scan logs from all connected terminals (FR-001) to check for an immediate match.

6. System generates a unique Claim ID and stores the claim with status "Received".

7. System displays a confirmation screen showing the Claim ID and expected next steps to the Passenger.

8. Use case ends successfully.

Alternate Flow — 4a: Invalid or Incomplete Claim Details

4a1. At step 4, the System detects a missing mandatory field or a malformed baggage tag.

4a2. System rejects the submission and highlights the specific field(s) in error with a corrective message.

4a3. Passenger corrects the highlighted details and resubmits the form, returning the flow to step 4.

4a4. If the Passenger abandons the form instead of correcting it, the registration is cancelled and no claim record is created.